

Venue Volume offline cue player

Schematic and build packet | Revision A | 3 October 2026

A self-contained venue playback box

Load a compiled cue sequence over local Ethernet or USB, then disconnect the uploading device. The box stores the show and generates DMX512 and Art-Net from its own clock.

Two isolated DMX outputs. Eight logical universes. Local microSD. Physical GO, HOLD and BLACKOUT. External 5 V power. No internet required for playback.

Design maturity

Engineering reference design with selected core components, pin-level schematic sheets, a purchasing list and implementation requirements. The hardware is unbuilt. Native ECAD, layout, Gerbers, firmware and physical tests remain development work.

Use at a venue

Connect directly to DMX distribution or Art-Net receivers. Connection through a console requires a documented input and merge policy for that exact desk. Channel playback does not import a native console show file.

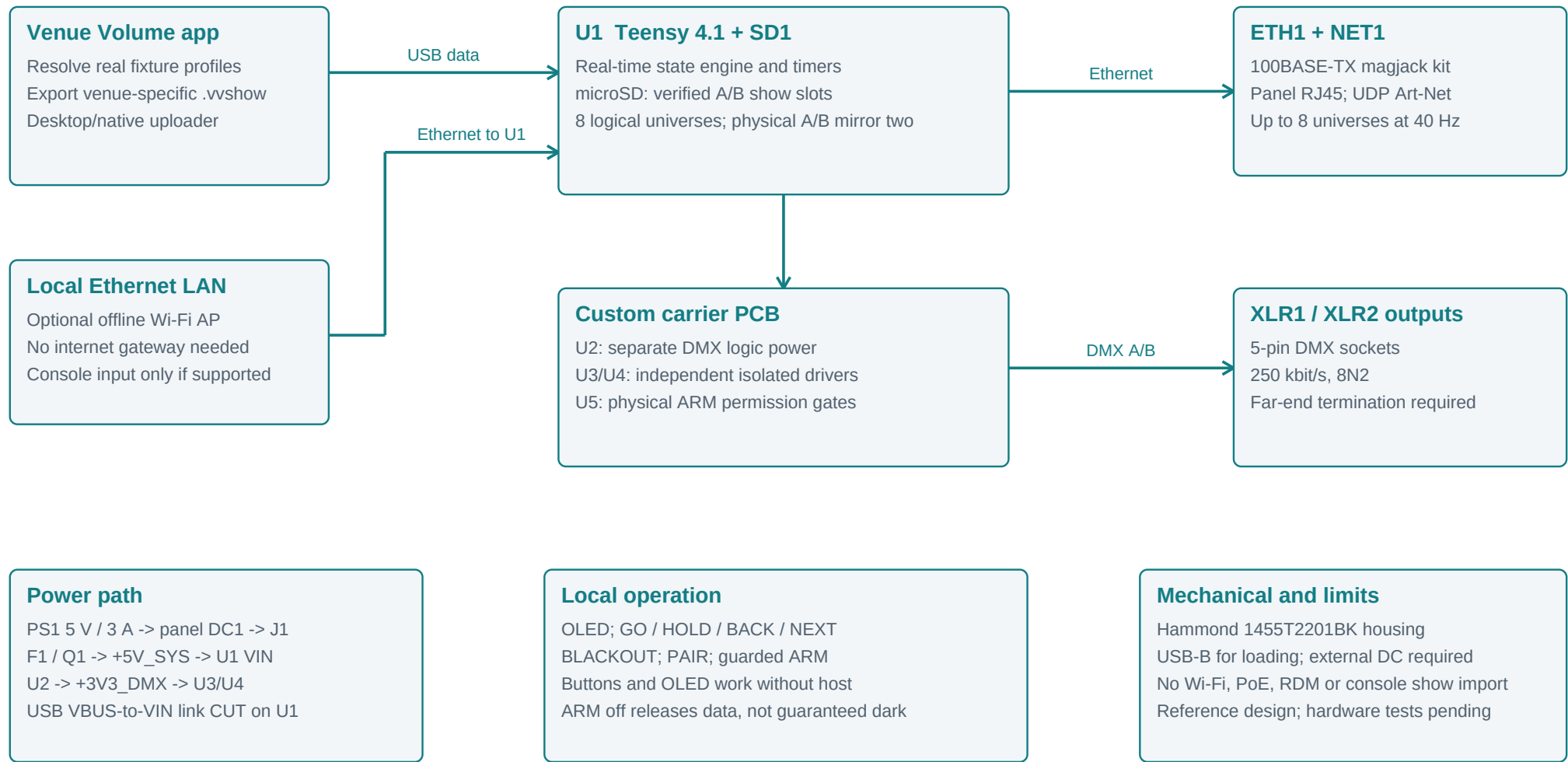
Packet contents

The next six landscape sheets show the system, power circuit, controller and control wiring, both isolated DMX circuits, and the enclosure arrangement. The following pages contain the product/build guide, electrical details, upload/runtime contract, manufacturing requirements, full BOM and source references.

Editable sources and the machine-readable reference connections live in hardware/cue-player. Documentation consistency checks are not electrical or manufacturing certification.

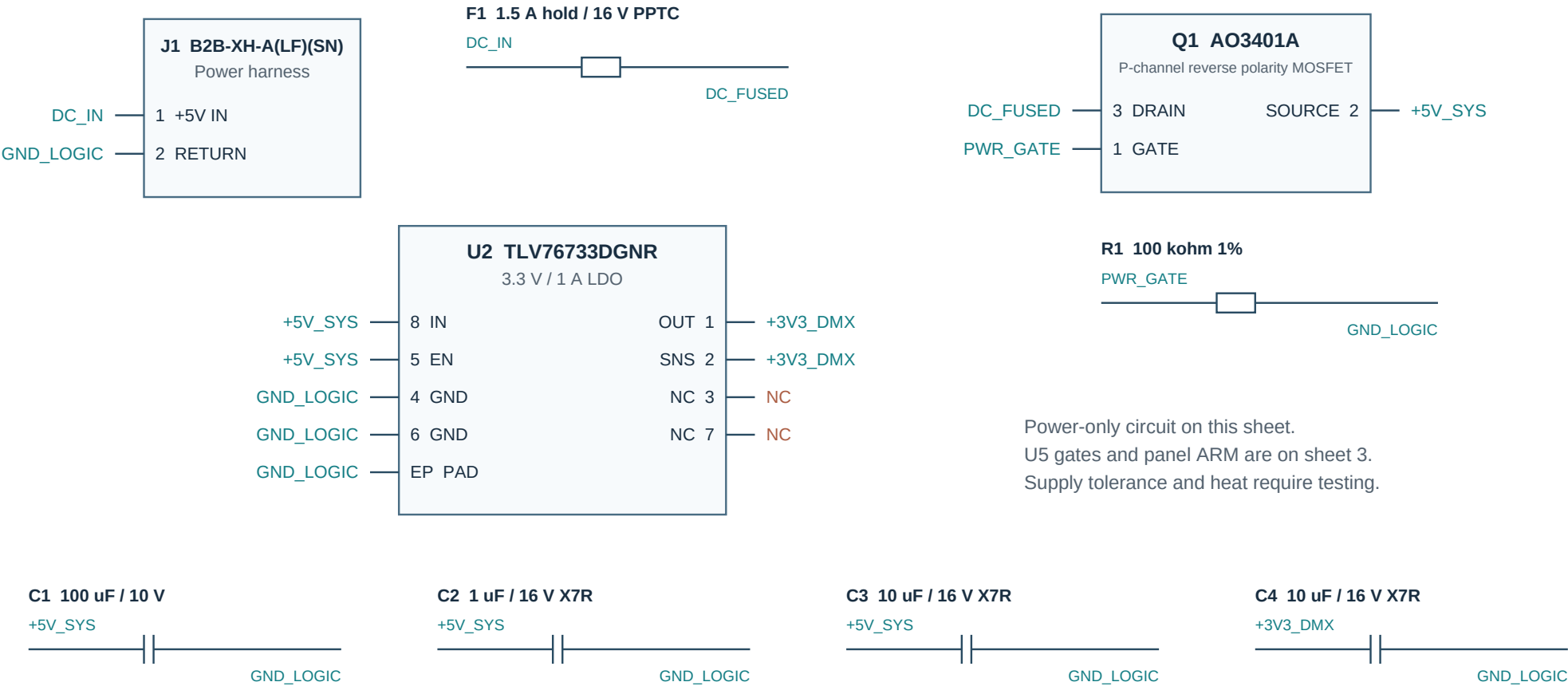
System interconnect

Store the show once. Local timing and output continue after the uploading client disconnects.



Power regulation and USB supply separation

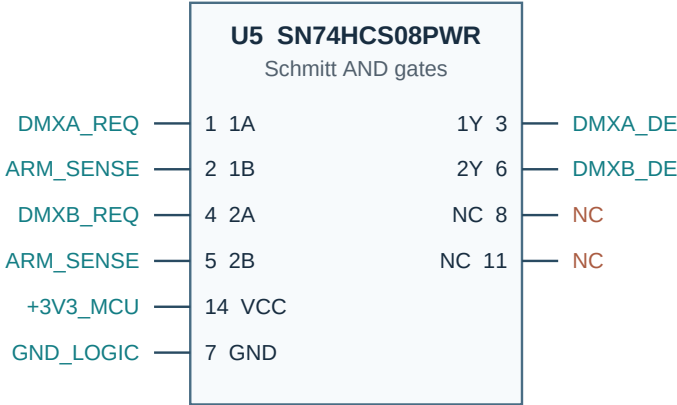
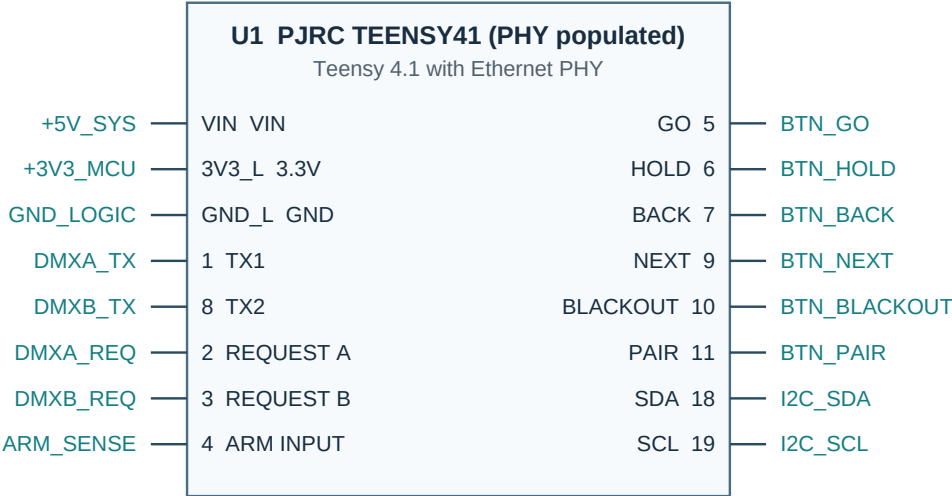
Logic ground and both DMX cable grounds remain separate. No USB-derived carrier power.



U1 VIN = +5V_SYS. U1 3.3 V output = +3V3_MCU. Never join +3V3_MCU and +3V3_DMXX.
C1 positive terminal faces +5V_SYS. Cut U1 VUSB/VIN link; retain USB VBUS at the module device port.

Controller, enable gates and panel connections

Module pins use Teensy labels. All unspecified GPIO are NC; full header mapping is in electrical.md.



U5 unused inputs 9/10/12/13 -> GND_LOGIC.
C5: 100 nF across U5 pins 14/7.
R4/R5: 10k requests to GND; R6/R7: 10k enables to GND.
R8/R9: 10k UART TX to GND. Initialize TX high before DE.

Panel harness J5 / JST-XH 12-way

1 GND; 2 +3V3_MCU; 3 GO_RAW; 4 HOLD_RAW; 5 BACK_RAW
6 NEXT_RAW; 7 BLACKOUT_RAW; 8 PAIR_RAW; 9 ARM_RAW
10 RUN_LED_A; 11 FAULT_LED_A; 12 GND
SW2-SW7: each RAW signal -> normally-open button -> GND
R20-R25: RAW -> 1k -> BTN; R30-R35: BTN -> 10k -> 3.3V
C30-C35: each BTN -> 100 nF -> GND. Debounce in firmware.
U1 pins 14/15 -> R16/R17 1k -> LED1/LED2 anodes; cathodes GND
U1 other GND and 3.3V header pins join corresponding module nets.

ARM, OLED and chassis

SW1: +3V3_MCU -> maintained switch -> ARM_RAW
R2: ARM_RAW -> 1k -> ARM_SENSE; R3: ARM_SENSE -> 10k -> GND
C7: ARM_SENSE -> 100 nF -> GND; MCU pin 4 reads ARM_SENSE
J4: 1 GND, 2 +3V3_MCU, 3 SDA, 4 SCL -> OLED1 (Adafruit 326)
C6: 100 nF at J4 supply; module supplies I2C pullups
R40: GND_LOGIC -> 0 ohm -> CHASSIS; J6 bonds enclosure
USB / Ethernet / XLR connector shells -> CHASSIS
No DMX isolated common joins this chassis or logic ground.

FB1: ISOA_RAW_3V3
-> bead -> ISOA_3V3

FB2: ISOA_CONV_GND
-> bead -> ISOA_GND

600 ohm at 100 MHz initial
candidate; final EMI review.

11/14 and 16/20 are NOT
one copper pour. Join only
through the ground bead.



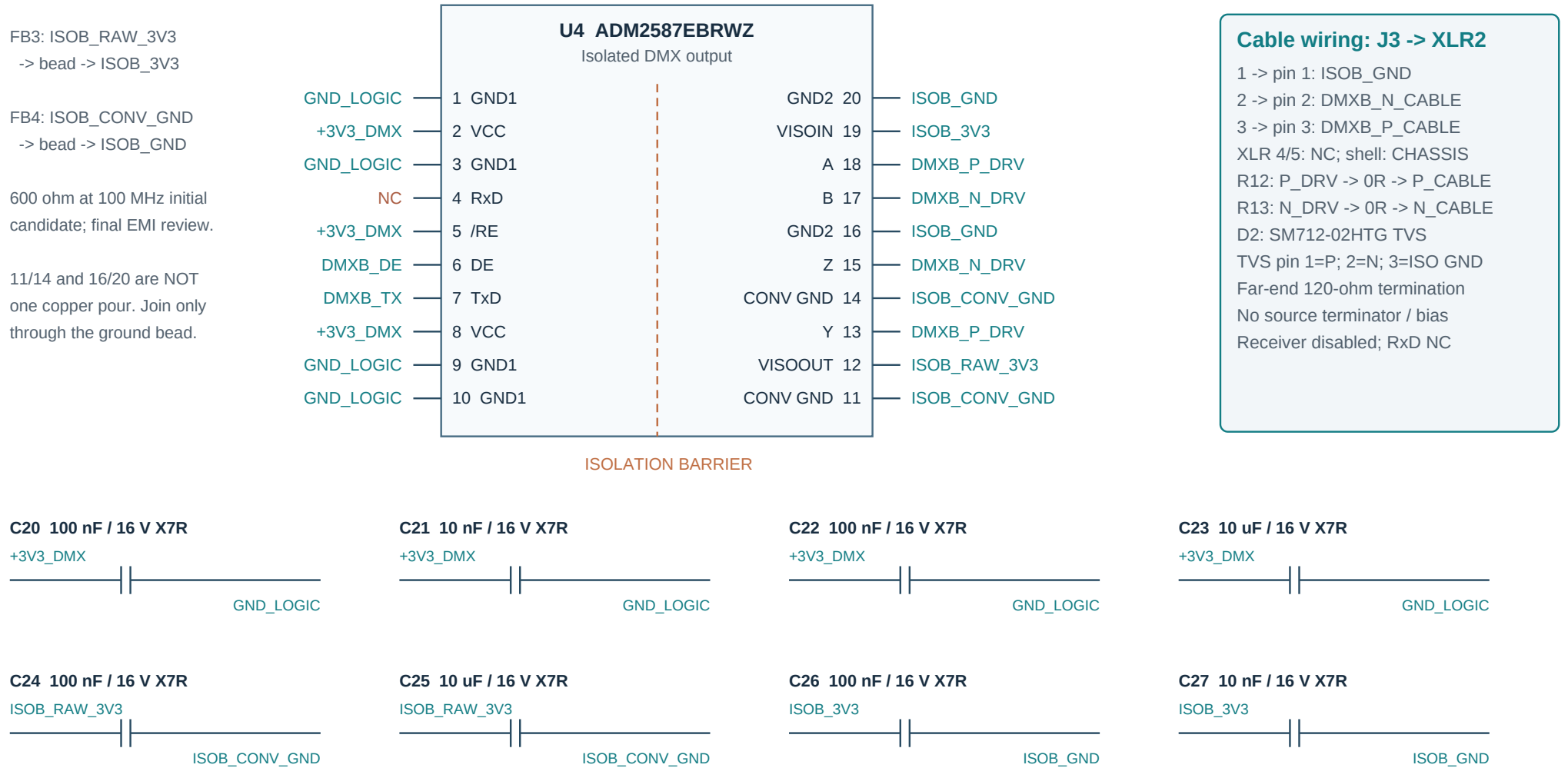
1 -> pin 1: ISOA_GND
2 -> pin 2: DMXA_N_CABLE
3 -> pin 3: DMXA_P_CABLE
XLR 4/5: NC; shell: CHASSIS
R10: P_DRV -> 0R -> P_CABLE
R11: N_DRV -> 0R -> N_CABLE
D1: SM712-02HTG TVS
TVS pin 1=P; 2=N; 3=ISO GND
Far-end 120-ohm termination
No source terminator / bias
Receiver disabled; RxD NC



REV A | 2026-10-03 | ENGINEERING REFERENCE - NOT FOR FABRICATION
Same net name = electrical connection. NC = deliberately unconnected. Read electrical.md and sources.md.

Isolated DMX output B

U4 is ADM2587EBRWZ. The ISOB cable domain is isolated from logic and from the other port.



Caps by pin pair: first two 2/1; next two 8/9; next two 12/11; last two 19/20. Place at each pair.

Enclosure and PCB placement allowance

Arrangement study only. Nominal dimensions and positions are not a machining or drill drawing.

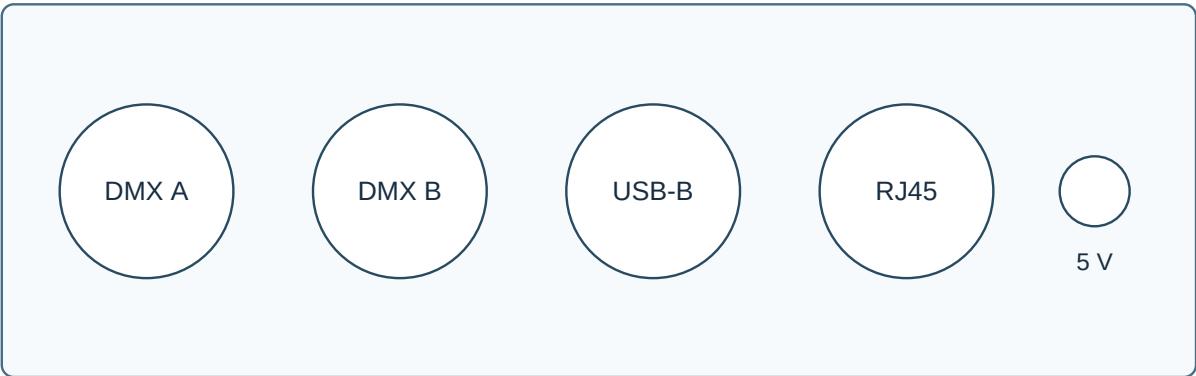
FRONT PANEL - 165 x 51.5 mm nominal



Carrier placement target

140 x 110 mm on insulating standoffs
U1 / SD in center; display harness forward
U3/U4 at rear, separate isolated islands
Rear XLR harnesses short and twisted
Ethernet kit on a retained bracket
Do not hang modules from their cables
Check actual internal envelope in CAD

REAR PANEL - 165 x 51.5 mm nominal



Before any panel is cut

Use manufacturer end-plate drawings
Check D-series flanges and screw holes
Select switch bodies and ARM guard
Check DC plug and RJ45 cable clearance
Secure internal USB and microSD
Test closed-enclosure temperatures
No production fit verification performed

Enclosure: Hammond 1455T2201BK, 220 mm depth. No mains supply or Wi-Fi radio inside.
This drawing communicates arrangement. Manufacturing requires native mechanical CAD and ECAD.

Venue Volume offline cue player

Revision A, engineering reference design, 2026-10-03. This packet specifies a small box that stores a venue-specific show, accepts uploads over local Ethernet or USB, and plays DMX512 and Art-Net without an app, computer, router uplink, or internet connection during the show. The initial upload can also be entirely offline if the show has already been exported.

This is a schematic and implementation specification for a prototype, not a fabricated or tested product. It includes component connections, a bill of materials, enclosure planning, firmware requirements, and a PCB release checklist. There is no routed PCB, Gerber set, implemented device firmware, or claim of console interoperability yet.

Design target

Item	Revision A proposal
Controller	PJRC/SparkFun Teensy 4.1 with Ethernet PHY, mounted on a custom carrier PCB
Local storage	Industrial/high-endurance 8-32 GB microSD in the controller's native SDIO socket
Physical outputs	Two independently isolated, output-only DMX512 universes on 5-pin XLR sockets
Network output	Up to eight logical universes over 100BASE-TX Art-Net; physical ports mirror two of these eight
Timing	40 complete frames/second; 250 kbit/s, 8N2 on each physical DMX port
Upload	Local Ethernet through a companion/native app, or USB device serial from a Mac/Windows/Linux companion
Front controls	GO, HOLD, BACK, NEXT, BLACKOUT, PAIR; guarded ARM toggle; OLED; run and fault LEDs
Power	External regulated 5 V supply, 3 A selected; USB does not power the box
Enclosure	Hammond 1455T2201BK, nominal 220 x 165 x 51.5 mm; custom front/rear panels
Show operation	Manual GO or locally timed follows; no cloud lease, periodic login, or time server needed to continue playback

Two physical outputs and eight network universes are explicit starting assumptions, not limits of the Venue Volume product. The USB host is assumed to be a computer until the target host is confirmed. Direct USB loading from visionOS or iPadOS is not promised. A Vision Pro can use the native app over a local Wi-Fi access point connected to the box's Ethernet LAN; that access point does not need internet. The enclosure contains no radio.

The module approach retains the controller's existing USB, memory, clock, Ethernet PHY, bootloader and SD circuitry. The custom PCB adds protected power, control connectors and two isolated line drivers. A later all-in-one MCU PCB is a separate redesign; do not copy this carrier's pin map into a bare processor layout.

Files in this packet

- `schematics/`: editable SVG sheets with named electrical nets and component pins; identical names connect across sheets.
- `connectivity.json`: reference connectivity, including explicit unconnected pins; module pins use the manufacturer's signal labels.
- `bom.json` and `bom.md`: component quantities, values, packages and procurement notes.
- `electrical.md`: power, isolation, processor pin assignment, wiring and circuit calculations.
- `firmware.md`: local upload, show compilation, playback and failure behavior.
- `manufacturing.md`: enclosure, PCB design rules, assembly and acceptance tests.

- `sources.md`: manufacturer and protocol references, including unverified details to resolve before PCB release.
- `build_packet.py`: reproduces diagrams, connectivity, BOM and the printable packet.
- Printable schematic and build packet.

How it connects at a venue

The normal route is box -> DMX fixtures or isolated splitter, or box -> Ethernet switch -> Art-Net nodes/fixtures. A console may share the network for monitoring or an explicitly configured handoff. The box performs the cue sequence itself.

If the venue wants the existing console to remain in control, the console must have a documented DMX input or an Art-Net input/merge function that accepts this source. Its universe mapping, channel ownership, merging and input licensing must be checked against that console and software version. A console's ability to output Art-Net does not establish that it can receive or merge it. Connecting two DMX outputs with a gender adapter or Y cable is not an input connection. Use a real input or an external merger with an agreed takeover policy.

DMX and Art-Net carry channel values, not a portable console cue database. This box does not install Venue Volume cues into an arbitrary desk's native show file. Triggering the desk's own cue list through OSC, MIDI or a vendor API is a different integration and is outside Revision A.

For simultaneous console and box control, assign distinct universes, use an explicit merger, or configure a documented exclusive handoff. HTP can make blackout ineffective when another source keeps intensities high; it is unsuitable as a general merge rule for pan, tilt, color wheels or control channels. Never infer exclusive ownership merely because discovery found no competing controller.

Operator workflow

1. In Venue Volume, select the venue, actual fixture personalities, patch, script and output routes. Compile and validate a complete playback bundle.
2. Power the box with ARM off. Connect a computer by USB, or join its local Ethernet network. Physically enable pairing for the first network upload.
3. Upload into the inactive storage slot. The box verifies integrity, format, resource limits and patch. Its display reports the show name, venue, revision and readiness.
4. Explicitly select the validated bundle while disarmed. Connect the approved lighting route. Check the displayed IP, universe mapping and output ownership at low intensity with the venue operator.
5. Turn ARM on and press GO. Disconnect the uploading computer and internet uplink. Cues and follows continue on the box's clock. GO and HOLD remain available on the front panel.
6. BLACKOUT sends the show's compiled dark state and latches until explicit restoration. ARM off releases transmission; fixtures may hold their last look when data stops. ARM is not a guaranteed blackout command.

The controls are for lighting playback. This revision does not implement safety-rated stops or allow pyro, hoists, lasers or other hazardous actuators to be armed through generic DMX channel data.

What exists in Venue Volume today

The repository already describes a venue-local bridge in `docs/03 Engineering/Service and Runtime Boundaries.md`, patch routing in F08, non-tracking cue states in F11, and continuity/command ownership in F14-F18. The design studio simulates those behaviors. The visionOS app explicitly labels its preview as having no physical output. Its `DMXPresets` currently validates only 1-16 preview channels, and `FixtureAiming` uses a pilot mapping rather than manufacturer personalities.

Therefore real fixture profile compilation, the portable show bundle, authenticated uploader, MCU playback runtime and physical output tests are new engineering work. Do not send preview arrays directly to venue equipment. This packet proposes those interfaces without claiming they are implemented.

Prototype purchasing budget

Allow approximately USD 350-650 for a first enclosed prototype, excluding engineering time, instruments, shipping, taxes and certification. This is a planning allowance, not a supplier quote. Major cost groups are the controller/SD/display, two isolated transceivers, carrier fabrication/assembly, panel connectors and harnesses, enclosure machining, and the power adapter. Obtain a fresh quantity-specific quote from `bom.md` after CAD review; generic panel parts still require a mechanical supplier selection.

Completion criteria

A bench prototype is ready for a supervised lighting test after the pin/footprint review, power and isolation checks, frame timing measurement, interrupted-upload tests and offline soak in `manufacturing.md` pass. Production ordering additionally needs native ECAD capture, ERC, a routed board, DRC, enclosure fit confirmation, firmware validation and emissions/immunity testing. These are required next development stages, not tests performed by this document.

Rebuilding and checking this packet

Run `python3 hardware/cue-player/build_packet.py --check` from the repository root for the dependency-free reference checks. PDF generation uses `reportlab`, `pypdf` and installed Liberation fonts: run `python3 hardware/cue-player/build_packet.py`. Use `--no-pdf` to regenerate only SVG, JSON and BOM files. Install dependencies in an environment inside the assigned worktree.

The checks cover 107 component/feature records, 50 named nets, six SVG documents, the isolated-domain separation, regulator-rail separation, selected pin assignments and DMX frame arithmetic. They do not run circuit simulation, ERC, DRC or hardware tests. Rendered PDF pages were visually reviewed for readability and clipping. The 19 test-pad records are copper features, not 19 extra purchased parts.

Electrical design

Revision A is a four-layer carrier for a Teensy 4.1 with its Ethernet PHY populated. Reference designators and named nets in the SVG sheets are authoritative within this packet. The generated connectivity file is a pin-level handoff specification, not an ECAD netlist or evidence of ERC.

Power and USB

PS1 is a regulated 5 V, 3 A external adapter, with a 5.5 mm outer / 2.1 mm inner barrel plug, center positive. Mean Well GST18A05-P1J is the selected reference; include the matching regional IEC mains lead. J1 is a locking two-position carrier power connector fed by a panel barrel socket. Specify a socket rated at least 3 A and positively verify plug fit and polarity.

J1.1 -> F1 -> Q1 drain (pin 3); Q1 source (pin 2) -> +5V_SYS. Q1 is an AO3401A P-channel MOSFET mounted with its body diode conducting from the input toward the load. Its gate (pin 1) goes through R1, 100 kohm, to GND_LOGIC. This orientation provides reverse-polarity protection. F1 is a 1.5 A hold, 16 V PPTC. Neither part protects against plugging in a higher-voltage adapter; label the inlet **5 V DC ONLY**.

C1, 100 uF/10 V electrolytic, and C2, 1 uF X7R/16 V, decouple +5V_SYS. Feed U1 VIN directly from this rail. Cut the manufacturer-designated VUSB-to-VIN link on U1 before connecting both external power and USB. Verify the open link and absence of backfeed on every unit. USB VBUS remains connected inside the module to the USB circuitry; it is not routed to the carrier power rail. A computer-only USB connection intentionally cannot start playback.

Use a Neutrik NAUSB-W feedthrough with USB-B facing outward and USB-A inward, plus a short USB-A to Micro-B data cable into U1's device port. This avoids an unverified custom USB-C/PD circuit. The app companion uses USB CDC; firmware updates use the module's service bootloader. Do not connect the USB host header to this inlet. Secure the internal Micro-B plug against vibration.

U2, TLV76733DGNR, supplies only the two transceivers' logic-side power as +3V3_DMX. Connect pins 8 IN and 5 EN to +5V_SYS; pins 1 OUT and 2 SNS to +3V3_DMX; pins 4, 6 and exposed pad to GND_LOGIC; leave pins 3 and 7 NC. C3 and C4 are 10 uF X7R/16 V at input/output. Preserve at least the regulator's required effective output capacitance after DC-bias derating. The exposed pad needs a ground copper area and thermal vias.

U1's own 3.3 V output is a separate net, +3V3_MCU. It powers only U5, the OLED and controls. Never connect it to +3V3_DMX or feed an external regulator into it. The external load target is under 60 mA, below PJRC's 250 mA recommendation. All GPIO is 3.3 V.

Power allowance

Load	Design allocation from 5 V
U1 including Ethernet PHY, CPU and SD bursts	350 mA
OLED, logic, controls and LEDs via U1 regulator	60 mA
Two ADM2587E channels via U2	300 mA
Margin for component variation	90 mA
Total allocated peak	800 mA, approximately 4 W

These allocations are not measured consumption. At 300 mA U2 dissipates approximately $(5.0 - 3.3) \times 0.30 = 0.51$ W. Check actual rail current, enclosure temperature and PPTC derating at the operating temperature target. The external 15 W supply adds margin, not permission to draw 3 A through a 1.5 A fuse. Test VIN at minimum supply voltage with cable, fuse and MOSFET losses.

Processor pin assignment

U1 pin label	Net	Function
VIN; all GND pins	+5V_SYS; GND_LOGIC	Module power
Both 3.3V header pins	+3V3_MCU	Output from module regulator
1 / TX1	DMXA_TX	Hardware UART1 TX to U3.7
8 / TX2	DMXB_TX	Hardware UART2 TX to U4.7
2	DMXA_REQ	Driver request into U5.1
3	DMXB_REQ	Driver request into U5.4
4	ARM_SENSE	Read the physical arm permit
5, 6, 7	BTN_GO, BTN_HOLD, BTN_BACK	Active-low buttons
9, 10, 11	BTN_NEXT, BTN_BLACKOUT, BTN_PAIR	Active-low buttons
14, 15	LED_RUN, LED_FAULT	LED drive through 1 kohm
18 / SDA; 19 / SCL	I2C_SDA; I2C_SCL	OLED, 100 kHz target
Dedicated 2x3 Ethernet pads	Module kit connection	Supplied keyed/orientation-marked cable and magjack kit
Onboard microSD	Native SDIO	No SPI pin assignment
Onboard Micro-B	USB device	Upload/service cable

Other GPIO, VUSB carrier pads, VBAT, On/Off, Program and USB host pads remain unconnected on the carrier. The onboard Program button is accessible through a recessed service opening. A real-time-clock coin cell is not required: sequence timing uses a monotonic timer. Do not use a wall clock to schedule fades.

For socket footprints, orient U1 with its USB connector at the top. The left 24-position row from top to bottom is GND, 0-12, 3.3V, 24-32. The right row is VIN, GND, 3.3V, 23-13, GND, 41-33. Validate row spacing and pad coordinates against the module drawing and an actual board before layout release. Signal labels in this packet do not imply an arbitrary KiCad symbol's pad numbering.

Physical enable gate

U5 is SN74HCS08PWR, a quad AND gate with Schmitt inputs. SW1 connects +3V3_MCU to ARM_RAW through the panel harness. R2, 1 kohm, connects ARM_RAW to ARM_SENSE; R3, 10 kohm, pulls ARM_SENSE low; C7, 100 nF, filters it to GND_LOGIC. The Schmitt input accommodates the slow switch edge.

Gate 1: pin 1 DMXA_REQ AND pin 2 ARM_SENSE -> pin 3 DMXA_DE. Gate 2: pin 4 DMXB_REQ AND pin 5 ARM_SENSE -> pin 6 DMXB_DE. Pin 14 is +3V3_MCU; pin 7 is GND_LOGIC; C5 is 100 nF. Unused inputs 9, 10, 12 and 13 go to GND_LOGIC; outputs 8 and 11 are NC. R4/R5 pull requests low and R6/R7 pull enables low, all 10 kohm. R8/R9 pull TX low, 10 kohm, so U3/U4 inputs never float during reset. Firmware sets TX high before asserting a request.

ARM off removes physical DMX drive regardless of the MCU request. Opening the switch takes roughly milliseconds to pass the RC/Schmitt threshold and can truncate a frame. Firmware also stops Art-Net on the observed disarm event; the gate is not in the Ethernet path. An internal MCU watchdog with a proposed 250 ms deadline resets a stalled runtime, with requests returned to inputs and pulled low. Validate that behavior on the selected bootloader and all reset paths. Loss of output can leave fixtures at their last value; neither the gate nor the watchdog provides a safety-rated blackout.

Isolated DMX channels

U3 and U4 are ADM2587EBRWZ, SOIC-20 wide body, 500 kbit/s devices with integrated isolated power. They are powered at 3.3 V on the logic side. The two external cable domains have separate grounds and isolated rails. Never join them to each other, USB ground, chassis, or GND_LOGIC. The isolation component rating alone does not certify the PCB or assembled product.

The following connections apply independently to U3 / port A and U4 / port B. See sheets 4 and 5 for the exact A/B reference designators.

IC pins	Connection
1, 3, 9, 10 GND1	GND_LOGIC
2, 8 VCC	+3V3_DMX
4 RxD	NC, receiver not used
5 /RE	+3V3_DMX, receiver disabled
6 DE	Corresponding gated enable
7 TxD	Corresponding UART TX
11, 14 converter ground	ISOx_CONV_GND, a local island
12 VISOOOUT	ISOx_RAW_3V3
13 Y, 18 A	DMXx_P_DRV
15 Z, 17 B	DMXx_N_DRV
16, 20 bus ground	ISOx_GND
19 VISOIN	ISOx_3V3

Two ferrites per port follow the manufacturer's Rev H arrangement: ISOx_RAW_3V3 -> FB -> ISOx_3V3, and ISOx_CONV_GND -> FB -> ISOx_GND. Do not directly short pins 11/14 to pins 16/20 around the ground ferrite. Use a high-frequency 0603 bead with adequate current and impedance, initially BLM18AG601SN1D; final impedance choice is subject to the EMI test. The ground ferrite connects only the two islands of that same isolated domain.

Each IC has eight bypass capacitors: 100 nF + 10 nF at pins 2/1; 100 nF + 10 uF at 8/9; 100 nF + 10 uF at 12/11; 100 nF + 10 nF at 19/20. Place each pair at its own pin pair even where the net names repeat. These are included in the BOM, not implied optional capacitors.

Each output goes through two 0-ohm 0805 tuning links to its cable connector; do not populate arbitrary series resistance without rechecking loaded output voltage. An SM712-02HTG TVS array connects its pins 1/2 to cable D+/D- and pin 3 to that port's ISOx_GND. This is a prototype transient network, not a demonstrated surge rating: the TVS clamp can exceed the transceiver's DC absolute maximum. Pulse coordination and system-level ESD/EFT/surge tests may require a revised protection stage.

Carrier J2/J3 pin 1 is isolated common, pin 2 D-, pin 3 D+. Wire these to matching pins 1/2/3 of the NC5FD-LX panel output sockets. XLR pins 4 and 5 are unconnected. XLR shells connect to metal chassis, separately from pin 1. Preserve the pair twist to the connector and keep these internal leads short. DMX512 cable is nominal 120 ohm; terminate only the far receiver end with a 120-ohm terminator for this output-only topology. No source-end parallel terminator or bias divider is fitted. Each external line is one daisy chain; use an isolated splitter for branches. RDM and DMX reception are not implemented by this circuit.

Control and display harnesses

J4 is a four-pin JST-SH/Qwiic connector: 1 GND_LOGIC, 2 +3V3_MCU, 3 SDA, 4 SCL. Use a standard cable to the current Adafruit 326 I2C/STEMMA QT OLED. Its I2C pullups are provided by that module; do not add a second pullup pair without measuring the combined resistance. Confirm 0x3C/0x3D during assembly. C6 is 100 nF at J4.

J5 is a 12-position JST-XH carrier connector: 1 ground, 2 +3V3_MCU, 3 GO_RAW, 4 HOLD_RAW, 5 BACK_RAW, 6 NEXT_RAW, 7 BLACKOUT_RAW, 8 PAIR_RAW, 9 ARM_RAW, 10 RUN_LED_A, 11 FAULT_LED_A, 12 ground. Each button shorts its raw input to ground. Each raw input has a 1-kohm series resistor to the MCU net; each MCU net has a 10-kohm pullup and 100-nF capacitor to ground. Debounce for 20 ms and require a release before a second GO; BLACKOUT gets highest service priority. LED anodes come from pins 14/15 through 1 kohm; cathodes return to panel ground. Limit cable length to 200 mm and keep it inside the enclosure. This is not an external remote-control port.

J6 is a single chassis bond tab. R40, 0 ohm, joins GND_LOGIC to CHASSIS at this one point in Revision A. Bond USB and Ethernet connector shells and XLR shells to chassis at their panels. Maintain all DMX isolation boundaries even if USB ground is earthed by a computer. Review additional internal shield/ground connections in the purchased Ethernet/USB modules during EMC work; this starting bond arrangement is not a claim of an optimized enclosure.

Upload and playback implementation

This is a proposed interface and runtime contract. No protocol service, compiler or firmware is implemented by the schematic packet. Hardware can operate offline only after this software is built and verified.

App integration

Add an Export to Cue Player flow that pins show, template, venue, fixture-profile, patch and script revisions. Resolve real manufacturer channel maps and complete fixture states before compilation. Reject unresolved profiles, overlapping patches, footprints extending beyond slot 512, unsupported modes, unsafe actuator categories and duplicate output ownership.

Use a small native desktop companion for the first USB and LAN uploader. It receives an exported bundle from the SaaS app or local disk. A native Venue Volume client can later implement the same LAN transport. An ordinary hosted browser is not assumed to have raw UDP, serial access or unrestricted local HTTPS access; origin, certificate and local-network-permission handling must be designed for each supported host. The current visionOS preview channel arrays are not a device export format.

Bundle proposal

Use a versioned `.vvshow` ZIP container with `manifest.json`, `states.bin`, `parameters.json` and `script.json`. Implement bounded streaming extraction, reject traversal and duplicate paths, cap uncompressed bytes and compression ratio, and allow only these payload entries. A prototype may use ZIP STORE only to bound decompressor complexity. Payload hashes are SHA-256 and are checked after receipt; hashes detect corruption but do not authenticate the sender.

Manifest field	Meaning and constraint
<code>formatVersion</code> , <code>minimumFirmware</code>	Exact format compatibility; reject unsupported versions
<code>showId</code> , <code>venueId</code> , <code>revision</code> , <code>displayName</code>	Immutable identity and operator-readable selection
<code>profileSetHash</code> , <code>patchHash</code>	Bind playback to the reviewed fixture setup
<code>universeCount</code>	1-8; all state records use the same universe order
<code>universeOrder</code>	Stable logical universe IDs, not implicit array labels
<code>routes</code>	Physical port A/B mapping plus Art-Net Port-Address and allowed destination configuration
<code>frameRate</code>	Revision A fixed at 40 Hz
<code>baselineState</code> , <code>darkState</code>	Valid complete-state record indexes
<code>payloads</code>	Exact byte lengths and SHA-256 of each allowed payload
<code>scriptHash</code>	Binds cue ordering, follows and fade policy

`states.bin` holds complete 512-byte, 8-bit channel arrays in manifest universe order, one record per resolved target/baseline/dark state. Record stride is $512 * \text{universeCount}$; reject any length mismatch. DMX START code is not stored in these arrays. A fixture's 16-bit parameter occupies two ordered slots but interpolates as one 16-bit value, then splits into coarse/fine bytes.

`parameters.json` describes per-slot behavior: 8-bit continuous, paired 16-bit continuous, discrete switch-at-start/end, dimmer master participation and blackout policy. Do not interpolate color-wheel slots, reset macros or other discrete control channels as intensity. Unused slots have explicit fixture/profile-approved values. A universal all-zero frame is not a safe replacement for a compiled dark state: pan/tilt, shutter and control channels can behave differently.

`script.json` identifies entries independently from cue IDs, with target-state index, fade duration, hold/follow time and terminal behavior. An entry with no follow waits for GO. END holds the final state. Follow time is measured from completion

of the preceding fade in this revision. Repeated references to the same cue remain different script entries. BACK/NEXT select a pending entry while held/disarmed; GO applies it. No implicit restart on select or reconnect.

Suggested initial limits: 1,000 cue entries, 8 universes, 16 MiB uncompressed bundle, 64 KiB manifest, 256 KiB script, and an explicitly bounded parameter table. At eight universes, 1,000 raw complete target states occupy 4,096,000 bytes before baseline/dark and metadata. Use one canonical state record per unique target where helpful. Admission must account for actual files, not this example alone.

Transport and storage

LAN: the companion discovers `_venuevolume._tcp` by mDNS, with an IP entry fallback. Proposed control/upload endpoint is HTTPS on TCP 443 with a per-device certificate/public-key fingerprint provisioned at assembly and stored by the companion. Physical PAIR while disarmed opens a 60-second enrollment window; display a one-time code. Bind the enrolled controller key to the device. Prototype private-key storage is not tamper-resistant. Do not expose a port forward or require a cloud token refresh for an active show. Pinning and device-key enrollment must work with no accurate wall clock or internet certificate lookup.

Proposed operations: GET `/v1/status`, POST `/v1/uploads`, chunked PUT `/v1/uploads/{id}`, POST `/v1/uploads/{id}/verify`, POST `/v1/bundles/{hash}/activate`, and ordered control commands. Activation requires disarmed state. Status reports received, verified, selected and active separately. Fixed request/buffer limits and upload pacing prevent TLS/file work from blocking output tasks.

USB: self-powered USB CDC device, desktop companion only in the initial compatibility target. Define a framed binary stream with magic, version, message type, request ID, payload length, payload and CRC32. Bound payloads to 4 KiB, ACK by request ID, support retry/resume by committed byte offset, and verify final SHA-256. Select a real VID/PID through an authorized allocation before product distribution; do not invent one or assume module development identifiers authorize a new product. Physical USB access is trusted for prototype upload while disarmed; it does not automatically arm outputs.

Store two show slots and redundant selection records on microSD. Write and verify the inactive slot, flush storage, then commit a generation-numbered/CRC-protected selection record. Keep the previous verified bundle until the new one has survived reload. Do not assume FAT rename or an SD flush makes removal of power atomic. Test power interruption at every stage. A partially written bundle is never selected; if both selection records are invalid, boot to recovery-required with outputs disabled.

Revision A permits uploads only while disarmed. Playback reads ahead into fixed RAM buffers; it does not mount the SD card as USB mass storage. No two hosts may write the filesystem. During a show, avoid filesystem metadata writes and flash/EEPROM writes in the timing path. Maintain bounded logs in RAM and save them when disarmed.

Runtime and output timing

Use MCU hardware timers, two independent UART transmit paths and DMA where supported. Network, SD and display activity run below frame deadlines; never bit-bang DMX from a general application loop. Preload current, target and dark states. Three eight-universe state banks require 12,288 bytes; allow more for interpolation starts, read-ahead, network buffers and TLS. One MiB module RAM is not wholly available to each memory/DMA class; validate the linker map, cache maintenance and actual worst-case allocation.

Physical output target is 250,000 bit/s, 8 data bits, no parity, 2 stop bits. Each frame contains a 120 us BREAK, 16 us mark-after-break, START code 0x00 and all 512 slots. A slot is $11 \text{ bits} / 250,000 = 44 \text{ us}$. Total active time is $120 + 16 + 513 * 44 = 22,708 \text{ us}$; at 40 Hz a 25,000 us period leaves 2,292 us of mark-before-break. Both universes can run in parallel. These are design targets; measure actual BREAK, MAB, baud error and jitter.

Art-Net uses UDP source and destination port 6454. Build ArtDmx with correct byte order, protocol version, a 512-byte channel payload and sequence 1-255 with wrap. Use 15-bit Port-Address mapping, $(\text{Net} \ll 8) | (\text{SubNet} \ll 4) | \text{Universe}$. Eight universes at 40 Hz carry 163,840 channel bytes/second before protocol/link overhead and destination fanout. Multiple unicast subscribers multiply traffic; cap the supported fanout and include it in the soak test.

Implement regular ArtPoll and subscriber tracking, then unicast ArtDmx to actual subscribers as required by the pinned Art-Net 4 specification. Do not label an ArtDmx broadcast implementation compliant with that specification. A manually configured destination for legacy non-discoverable equipment is an explicitly tested compatibility mode. ArtSync is optional

and enabled only for destinations known to support it. Do not assume UDP delivery acknowledges a fixture's actual light state.

Current Art-Net revision 1.4dp deprecates Port-Address zero and specifies 1-32767. Default logical Universe 1 to wire 0:0:1 (address 1). The existing F08 example of 0:0:0 remains a selectable, warned legacy mapping; it must not silently shift an existing show. Display logical ID, Net/SubNet/Universe and numeric address together. sACN is a possible future firmware transport on the same Ethernet hardware, not a Revision A feature.

Use a locally configured static lighting-network address or DHCP with displayed fallback and conflict handling. Commission a direct link over USB first if no DHCP server exists. Persist the reviewed network profile. Document the Art-Net specification's default addressing convention when implementing factory settings; private-address venue networks and subnet masks are explicitly configured installations. Do not infer a venue subnet or send unbounded discovery outside it. Art-Net needs a LAN, not an internet gateway. Discovery packets are unauthenticated protocol traffic and must never be accepted as permission to arm or install a show.

State and failure behavior

Event/state	Required action
Boot/reset/brownout	Requests low; no ArtDmx; validate selected bundle; require ARM off then on and explicit GO
Ready/disarmed	Inspect and upload; outputs disabled
ARM switch on	Permit ready state; never start a sequence from the switch alone
GO	Apply pending entry once; start a new transition/follow generation
HOLD	Freeze transition progress and cancel pending follows; continue transmitting the frozen look
BLACKOUT	Latch compiled dark output; cancel follows; require explicit restore of saved state, then GO to resume
App, USB or internet loss	Continue the already armed local run and physical controls
Ethernet loss	Continue physical DMX; show network fault; do not replay missed network cues on reconnect
Ethernet recovery	Discover/validate destinations and send current state; do not rewind the script
SD read error	Hold the last complete frame, cancel follows, show fault; dark state remains in RAM
Watchdog reset	Output releases and recovery-required on reboot; venue fixtures' loss behavior remains external
End of script	Hold final complete look until explicit operator action

BLACKOUT restoration must be a deliberate local menu action, distinct from an accidental second button edge. Restoring re-applies the saved frozen look; the operator separately resumes progression. DARK output cannot overcome another controller in an HTP merger. State names and interruption rules must agree with F11 and F17 before app implementation.

Remote commands carry command ID, session ID, selected bundle hash, operator lease ID, monotonically increasing sequence and playback generation. Deduplicate retries; reject stale sessions and late follows. A lease arbitrates remote commands locally; expiration does not stop an authorized autonomous sequence. Reconnection reads state first. Diagnostics distinguish a commanded value, a transmitted packet and any actual receiver feedback.

PCB and enclosure build requirements

Deliverables before ordering a PCB

Capture the supplied pin-level design in native ECAD, assign exact symbols/footprints and run electrical rules checking. Cross-check every IC pin and every module/header orientation against the source datasheets and physical samples. Generate a routed PCB, passing design-rule report, stack-up, copper/soldermask/silkscreen Gerbers, drill files, IPC-356 netlist if supported, BOM, pick-and-place data, assembly drawings, stencil and fabrication notes. None of those manufacturing files are supplied or implied by the SVG drawings.

Revision A's carrier envelope target is 140 x 110 mm on insulated M3 standoffs inside the selected enclosure. This is a placement allowance, not a final board outline or a claim it fits the enclosure rails. Confirm internal height, panel-to-board distance, SD access, plug bend radii and mounting positions in mechanical CAD before fixing the outline. A carrier is the only new PCB; controller, Ethernet magjack and OLED are bought as complete modules.

Board construction and placement

Proposed construction: four-layer FR-4, 1.6 mm, 1 oz outer copper, lead-free finish. Obtain the fabricator's stack-up before impedance calculations. L1 carries components/signals; L2 ground; L3 power/ground; L4 low-speed signals. Split every layer at both DMX isolation boundaries. Create separate same-port converter and bus ground areas joined only through the specified ferrite. Never pour the main ground plane under an isolated island.

Reserve at least 8 mm copper clearance between the logic domain and each isolated cable domain, and between cable domains, wherever practicable; the isolator footprint itself must preserve the package's certified barrier dimensions. Treat this as a conservative layout target, not a declared mains insulation class. Keep mounting metal, connector shells, test pads and silkscreen solder bridges out of the barriers. No added capacitor across an isolation barrier is authorized by this reference design. Any EMC stitching-capacitor change needs a reviewed insulation design.

Put transceivers, beads and bypass capacitors at the rear edge near their XLR harnesses. Follow the ADM2587E Rev H capacitor and island arrangement, including the isolated ground ferrite. Do not route display clocks through that area. Give U2 a real thermal plane and vias. Use adequate power trace/plane width for at least 1 A with the chosen stack-up; the fuse and connector ratings must also pass thermal review.

Keep UART traces short with a continuous logic return. USB high-speed pairs and Ethernet analog pairs remain in the purchased modules/cables. Do not route Ethernet PHY pairs through generic carrier headers or long unshielded harnesses. Use the supplied Ethernet kit cable and orientation. A short internal CAT5e patch cable connects the kit RJ45 to a shielded panel RJ45 feedthrough. Secure the kit mechanically; it cannot hang from the ribbon cable. No PoE circuitry is included.

Provide labeled test points for 5V_SYS, 3V3_MCU, 3V3_DMXX, GND_LOGIC, both requests/enables/TX signals, and each isolated rail/common/output pair. Each isolated test group stays within its own domain. Debug instruments with common earth leads can defeat isolation; use differential/isolated measurement for cross-domain measurements.

Enclosure and harness set

Use Hammond 1455T2201BK as the initial housing. Nominal outside dimensions are 220 x 165 x 51.5 mm. Its front and rear end plates need machining and durable labels. The schematic packet includes an arrangement drawing only; obtain the actual panel and connector drawings before cutting. Do not manufacture from a scaled screenshot.

Suggested front placement, measured from the nominal panel's left/bottom: OLED around x=29 mm, y=26 mm; run/fault indicators around x=53 mm; six 12-mm momentary buttons in two rows at x=72, 94, 116 mm and y=16, 36 mm; guarded ARM toggle at x=146 mm, y=26 mm. Choose switch bodies and guard envelopes before finalizing these centers.

Suggested rear placement: DMX A, DMX B, USB and RJ45 centered near x=20, 55, 90, 125 mm at mid-height; power inlet near x=151 mm. Neutrik D-series mounting holes and flanges extend outside the circular cutout and must be checked for clashes. The power plug strain relief must clear the Ethernet plug. If necessary move power to a side panel and keep it clear of the PCB.

Assembly includes: six low-current SPST-NO momentary panel switches; one maintained ARM switch/guard; green and red LEDs with bezels; two NC5FD-LX sockets; NAUSB-W feedthrough; shielded RJ45 feedthrough; insulated center-positive barrel inlet; OLED bezel/window; all mating connector housings/crimps; two short twisted DMX harnesses; 12-way controls harness; OLED cable; internal USB and CAT5e cables; Ethernet kit ribbon; M3 standoffs/screws/lock washers; module brackets; SD retention/service cover; non-slip feet; labels and a chassis bond lead. Exact commodity/mechanical SKUs are procurement choices with the electrical requirements listed in the BOM. Do not omit them from a build quotation.

A removable service panel gives access to the microSD and Program button. The card is not hot-swappable during a show. Restrain all internal cables, avoid sharp panel edges, and retain the controller with a bracket in addition to its sockets. Do not pot the isolation area or add conductive mounting hardware across it. Ventilation and a small controller heatsink can be added if thermal testing requires them; they are not substitutes for the regulator thermal design.

Assembly sequence

1. Inspect bare boards against the fabrication drawing. Assemble and inspect power components first, without U1 or DMX modules populated. Check polarity, exposed pad soldering and shorts.
2. Apply current-limited 5 V. Check +5V_SYS and +3V3_DMX. Remove power and populate U3/U4/U5 and all passives/connectors.
3. Cut and inspect U1's VUSB-VIN link, fit headers and sockets, and mechanically retain the module. Install the Ethernet kit following its assembly guide. Install SD and OLED.
4. Reapply current-limited power with ARM off; confirm rails, gate states and isolated voltages. Install firmware through the service USB path. Never enable outputs with unreviewed test channel values.
5. Build and continuity-test harnesses off-board, including XLR pin numbers rather than visual left/right assumptions. Install panels, strain reliefs and chassis bonds.
6. Perform the following acceptance tests. Record hardware serial, board revision, firmware hash, card type and all instruments/settings with the results.

Prototype acceptance tests

Test	Required evidence before venue use
Power	Current and rail voltage at idle, network/SD peaks and both loaded outputs; no brownout within the reviewed supply range
USB coexistence	No carrier-to-host VBUS backfeed with external power; plugging/unplugging USB does not restart or pause an armed show
Isolation	No DC continuity from either XLR signal common to logic/chassis or the other port; formal insulation test procedure reviewed separately
Physical enable	ARM off and reset disable both drivers; no unsolicited ArtDmx before explicit GO; GPIO/rail sequencing checked with a scope
DMX electrical	Correct XLR polarity and loaded differential levels; far-end 120-ohm termination; 32-unit-load equivalent and representative cable lengths
DMX timing	Capture both ports: baud, two stop bits, BREAK, MAB, all 512 slots, 25 ms period; target period jitter within +/-100 us under admitted load
Art-Net	Packet capture confirms ports, byte order, sequence wrap, mapping and subscriber unicast; test known node plus intended console if used
Upload/storage	Corrupt, oversized and wrong-version bundles rejected; power removal throughout upload/selection never chooses partial data
Offline operation	At least 24 hours with WAN removed and uploader disconnected, both DMX ports and eight network universes active; no skipped/duplicated transitions

Test	Required evidence before venue use
Control semantics	Button bounce, double GO, HOLD, BLACKOUT, restore, follows, restarts, stale remote commands and lease expiration match contract
Failure handling	Pull Ethernet, remove/fault SD, force watchdog reset and vary power; measure actual fixture hold/dark behavior and display errors
Mechanical/thermal	Enclosure closed at proposed 0-40 C ambient target; touch/component temperatures within limits; cable retention and panel fit verified
EMC	Pre-scan emissions and immunity with the real metal enclosure and attached cables; revise layout/protection if required

The jitter, soak duration and ambient range are proposed acceptance targets, not published product specifications. A JSON consistency check cannot establish any row in this table. Production release requires measured results, supply-chain review, regulatory assessment for the intended markets and a documented factory test fixture.

Development order

First build a module-based bench unit and verify one real fixture with a fixed known channel pattern. Then implement the compiler/uploader and local state engine, verify both physical outputs and an Art-Net receiver, and exercise all offline failure cases. Capture/review the carrier in ECAD before ordering five assembled boards. Fit the panels and run the closed-enclosure tests before a supervised venue trial. Add console-specific adapters only after documenting the target desk's supported input and takeover behavior.

Bill of materials

Quantities are for one complete Revision A box. Electronic references correspond to the pin-level sheets and connectivity.json. All resistors are at least 0.1 W unless stated; 0-ohm links must carry the relevant fault/test current. Capacitor ratings are minimums and effective capacitance must be verified. This is a reference purchasing list, not an assembly-house release BOM.

Core semiconductor MPNs are selected. Specification-controlled passives and commodity/mechanical items need exact vendor SKUs and footprints before purchase. Optional external LAN equipment has quantity zero and is not included in the base assembly.

Carrier electronics

References	Qty	Component and package	Part or selection requirement	Notes
U1	1	Teensy 4.1 with Ethernet PHY; Module / two 1x24 rows	PJRC TEENSY41 (PHY populated)	Cut VUSB-VIN link. USB device, SDIO and Ethernet kit remain on module. Unused GPIO NC.
U2	1	3.3 V / 1 A LDO; DGN HVSSOP-8 exposed pad	TLV76733DGNR	SNS at OUT; EP to thermal ground plane.
U5	1	Schmitt AND gates; TSSOP-14	SN74HCS08PWR	-
F1	1	1.5 A hold / 16 V PPTC; 1812	1812L150/16DR	Check ambient derating and fault clearing.
Q1	1	P-channel reverse polarity MOSFET; SOT-23	AO3401A	Pin 1 gate, 2 source, 3 drain. Drain faces incoming supply.
R1	1	100 kohm 1%; 0603	Specification controlled	-
C1	1	100 uF / 10 V; Radial electrolytic	Specification controlled	Positive terminal pin 1; reserve exact lead pitch after procurement.
C2	1	1 uF / 16 V X7R; 0603	Specification controlled	-
C3	1	10 uF / 16 V X7R; 0805	Specification controlled	-
C4	1	10 uF / 16 V X7R; 0805	Specification controlled	Check effective capacitance after bias derating.
C5, C6, C7, C30, C31, C32, C33, C34, C35	9	100 nF / 16 V X7R; 0603	Specification controlled	-
R2, R20, R21, R22, R23, R24, R25, R16, R17	9	1 kohm 1%; 0603	Specification controlled	-
R3, R4, R5, R6, R7, R8, R9, R30, R31, R32, R33, R34, R35	13	10 kohm 1%; 0603	Specification controlled	-
U3, U4	2	Isolated DMX output; SOIC-20 wide body	ADM2587EBRWZ	Output only. /RE high, RxD NC. Separate isolated supply per port.
FB1, FB2, FB3, FB4	4	600 ohm at 100 MHz ferrite; 0603	BLM18AG601SN1D	Initial EMI candidate; verify high-frequency curve, current and DCR before release.
C10, C20	2	100 nF / 16 V X7R; 0603	Specification controlled	IC pins 2/1
C11, C21	2	10 nF / 16 V X7R; 0603	Specification controlled	IC pins 2/1

References	Qty	Component and package	Part or selection requirement	Notes
C12, C22	2	100 nF / 16 V X7R; 0603	Specification controlled	IC pins 8/9
C13, C23	2	10 uF / 16 V X7R; 0805	Specification controlled	IC pins 8/9
C14, C24	2	100 nF / 16 V X7R; 0603	Specification controlled	IC pins 12/11
C15, C25	2	10 uF / 16 V X7R; 0805	Specification controlled	IC pins 12/11
C16, C26	2	100 nF / 16 V X7R; 0603	Specification controlled	IC pins 19/20
C17, C27	2	10 nF / 16 V X7R; 0603	Specification controlled	IC pins 19/20
R10, R11, R12, R13	4	0 ohm; 0805	Specification controlled	Tuning link. Default 0 ohm; added resistance requires loaded DMX tests.
D1, D2	2	RS-485 transient array; SOT-23	SM712-02HTG	Protection coordination requires bench validation; no claimed surge rating.
J2, J3	2	DMX harness; JST-XH 3-pin 2.5 mm	B3B-XH-A(LF)(SN)	-
J1	1	Power harness; JST-XH 2-pin 2.5 mm	B2B-XH-A(LF)(SN)	-
J4	1	OLED interface; JST-SH 4-pin 1 mm	SM04B-SRSS-TB(LF)(SN)	-
J5	1	Panel controls harness; JST-XH 12-pin 2.5 mm	B12B-XH-A(LF)(SN)	-
J6	1	Chassis bond point; M3 ring lug / bond pad	Specification controlled	-
R40	1	0 ohm; 0805	Specification controlled	Single initial logic/chassis bond; never bridge an isolated DMX common.

Panel components

References	Qty	Component and package	Part or selection requirement	Notes
XLR1, XLR2	2	5-pin DMX output socket; Panel D series	NC5FD-LX	-
SW2	1	GO momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.
SW3	1	HOLD momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.
SW4	1	BACK momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.
SW5	1	NEXT momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.
SW6	1	BLACKOUT momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.

References	Qty	Component and package	Part or selection requirement	Notes
SW7	1	PAIR momentary switch; 12 mm panel; low-current gold contacts	Specification controlled	SPST-NO; final mechanical SKU and button color selected before panel machining.
SW1	1	ARM maintained switch + guard; Panel SPST	Specification controlled	Low-current gold contacts. Guard envelope checked in panel CAD.
LED1	1	green panel LED; 3 mm LED + insulating bezel	Specification controlled	-
LED2	1	red panel LED; 3 mm LED + insulating bezel	Specification controlled	-
OLED1	1	128x64 I2C OLED; 29.2 x 26.7 mm module	Adafruit 326 current STEMMA QT version	Module supplies I2C pullups; verify address.

Test features

References	Qty	Component and package	Part or selection requirement	Notes
TP1	1	+5V_SYS; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP2	1	+3V3_MCU; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP3	1	+3V3_DMXX; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP4	1	GND_LOGIC; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP5	1	ARM_SENSE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP6	1	DMXA_REQ; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP7	1	DMXB_REQ; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP8	1	DMXA_DE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP9	1	DMXB_DE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP10	1	DMXA_TX; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP11	1	DMXB_TX; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP12	1	ISOA_3V3; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP13	1	ISOA_GND; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP14	1	DMXA_P_CABLE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.

References	Qty	Component and package	Part or selection requirement	Notes
TP15	1	DMXA_N_CABLE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP16	1	ISOB_3V3; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP17	1	ISOB_GND; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP18	1	DMXB_P_CABLE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.
TP19	1	DMXB_N_CABLE; 1.5 mm exposed test pad	Specification controlled	Copper feature; no purchased component.

Modules hardware cables and accessories

References	Qty	Component and package	Part or selection requirement	Notes
ETH1	1	Ethernet kit for Teensy 4.1; Assembly / accessory	PJRC ETHERNET_KIT	Includes magjack, PCB, capacitor, 2x3 headers and ribbon cable; assemble vendor kit.
SD1	1	Industrial/high-endurance 8-32 GB microSD; Assembly / accessory	Qualified supplier SKU required	FAT32 initial target; qualify exact model for power interruption/read latency.
PS1	1	5 V 3 A external adapter; Assembly / accessory	Mean Well GST18A05-P1J	Add regional IEC mains lead; no mains wiring in box.
USB1	1	Reversible USB A/B panel feedthrough; Assembly / accessory	Neutrik NAUSB-W	B outward, A inward; bond shell to chassis.
NET1	1	Shielded RJ45 panel coupler, CAT5e or better; Assembly / accessory	Mechanical SKU to select	Panel grounded shell; internal short patch cable to Ethernet kit; no PoE.
DC1	1	Insulated panel barrel socket, 5.5/2.1 mm, >=3 A; Assembly / accessory	Mechanical SKU to select	Center positive; sleeve to logic ground; two-wire harness to J1.
CASE1	1	Extruded aluminum enclosure; Assembly / accessory	Hammond 1455T2201BK	220 x 165 x 51.5 mm nominal; use custom machined end panels.
PCB1	1	Custom 4-layer carrier, 140 x 110 mm allowance; Assembly / accessory	Native ECAD and routed PCB required	Not supplied as fabrication-ready artwork.
SOCKET	2	1x24 female 2.54 mm module socket; Assembly / accessory	PJRC 24-pin socket or dimensional equivalent	Check controller header lengths and retention bracket.
HEADER	2	1x24 male 2.54 mm header; Assembly / accessory	PJRC 24-pin header or dimensional equivalent	For bare Teensy module; omit if purchased already fitted.
HARNESS1	1	J1 power mating connector and leads; Assembly / accessory	XHP-2 + 2 x SXH contacts	22 AWG stranded; verify crimp tooling and pin 1.
HARNESS2	2	J2/J3 DMX mating connectors and leads; Assembly / accessory	XHP-3 + 3 x SXH contacts per harness	Short 120-ohm twisted pair + isolated common; heatshrink at XLR.
HARNESS3	1	J5 12-way panel harness; Assembly / accessory	XHP-12 + 12 x SXH contacts	26 AWG signal wiring; <=200 mm; continuity test each switch.

References	Qty	Component and package	Part or selection requirement	Notes
CABLE1	1	JST-SH/Qwiic 4-wire OLED cable; Assembly / accessory	Standard keyed cable, <=150 mm	Verify pinout; use assembled cable, no hand-swapped wire colors.
CABLE2	1	Internal USB-A male to Micro-B data cable; Assembly / accessory	Short, shielded, <=200 mm	Secure both ends; no charge-only cable.
CABLE3	1	Internal CAT5e patch cable; Assembly / accessory	Short shielded patch, <=200 mm	Kit magjack to panel coupler.
CABLE4	1	Host USB cable; Assembly / accessory	USB-A/C host to USB-B device data cable	Host-dependent cable; supplied as accessory.
CABLE5	1	External CAT5e network cable; Assembly / accessory	Shielding per venue installation	Local Ethernet connection.
CABLE6	2	External DMX512 5-pin cable; Assembly / accessory	120-ohm DMX cable	One for each physical output as required.
TERM	2	120-ohm far-end XLR DMX terminator; Assembly / accessory	5-pin, 0.25 W or greater resistor	External accessory, no source-end shunt termination.
MECH	1	Mechanical hardware set; Assembly / accessory	Procure to reviewed mechanical CAD	Insulating M3 standoffs, screws, washers, module brackets, feet, OLED window, SD cover, strain reliefs, chassis lead and labels.
LAN	0	Optional offline Wi-Fi AP / Ethernet switch; Assembly / accessory	Venue-dependent external equipment	Only needed to join wireless app clients or multiple network nodes; internet not required.

Design sources and verification status

Sources were checked on 2026-10-03. URLs below identify the primary manufacturer or protocol owner. Component choices and calculations are engineering proposals derived from these sources; availability, footprints, firmware behavior and completed-system ratings are not inferred from a part listing.

Source	Use in this design
PJRC Teensy 4.1	MCU module, native USB/SD/Ethernet facilities, VIN/VUSB separation and 3.3 V external-current limit
PJRC pinout card	UART and I2C pin assignments and header signal labels
PJRC Ethernet kit	Complete magjack kit, cable and assembly, retaining the vendor's analog interface
Analog Devices ADM2582E/ADM2587E Rev H	U3/U4 pins, isolated power, bypass networks and Figure 35 ferrite/island arrangement
TI TLV767 Rev D	U2 fixed 3.3 V DGN package, SNS and thermal-pad connections
TI SN74HCS08 Rev C	U5 pinout, AND function and Schmitt-input behavior
AOS AO3401A Rev 3.1	Q1 P-channel MOSFET package and ratings
Littelfuse 1812L family	F1 1812L150/16DR electrical ratings
Littelfuse SM712	Candidate RS-485 TVS network; clamp coordination remains a test item
Murata ferrite selection guidance	BLM18AG601SN1 family and frequency-dependent impedance; candidate requires final EMI qualification
Adafruit 326	Current I2C OLED module, outline and wiring; old SPI variants must not be substituted blindly
Neutrik NC5FD-LX	Five-contact panel output socket and panel drawing
Neutrik NAUSB-W	Reversible USB A/B panel feedthrough
Hammond 1455T2201BK	Selected enclosure dimensions and mechanical drawing
Mean Well GST18A	Selected external GST18A05-P1J supply and plug option
Artistic Licence Art-Net 4, release 1.4, revision 1.4dp	Addressing, ArtDmx, unicast subscriptions, discovery, sequence and implementation credit
ESTA published standards index	Current DMX512-A baseline is ANSI E1.11-2024; full standard must be reviewed before release
Infineon AN56199 DMX transmitter application note	Historical DMX timing background; the application note is marked obsolete and does not replace the current normative standard

The ESTA full E1.11-2024 PDF endpoint returned an access error during research. The index establishes the current edition; this packet does not claim a clause-by-clause compliance review. Selected 120-us BREAK and 16-us MAB targets provide margin over historical transmitter timing minima, but must be checked against the current purchased/downloaded standard and measured hardware.

Art-Net product release requires the protocol owner's specified credit and an assigned OEM code. Include the required credit in the eventual user guide: "Art-Net(TM) Designed by and Copyright Artistic Licence" (use the trademark symbol in the final branded guide). Do not reuse another product's OEM code. A USB VID/PID allocation is an independent release requirement.

Unresolved before manufacturing: target console and software version; USB host platform beyond desktop; exact generic panel switch, inlet and RJ45 feedthrough SKUs; ferrite impedance selection and current derating; module footprint measurements; cable clearances; power/thermal margins; complete-system ESD/EMI behavior; native ECAD review. The core MCU, regulator, gate and isolator connections are specified here, while these remaining items are expressly not validated.